

Supplementary Figures

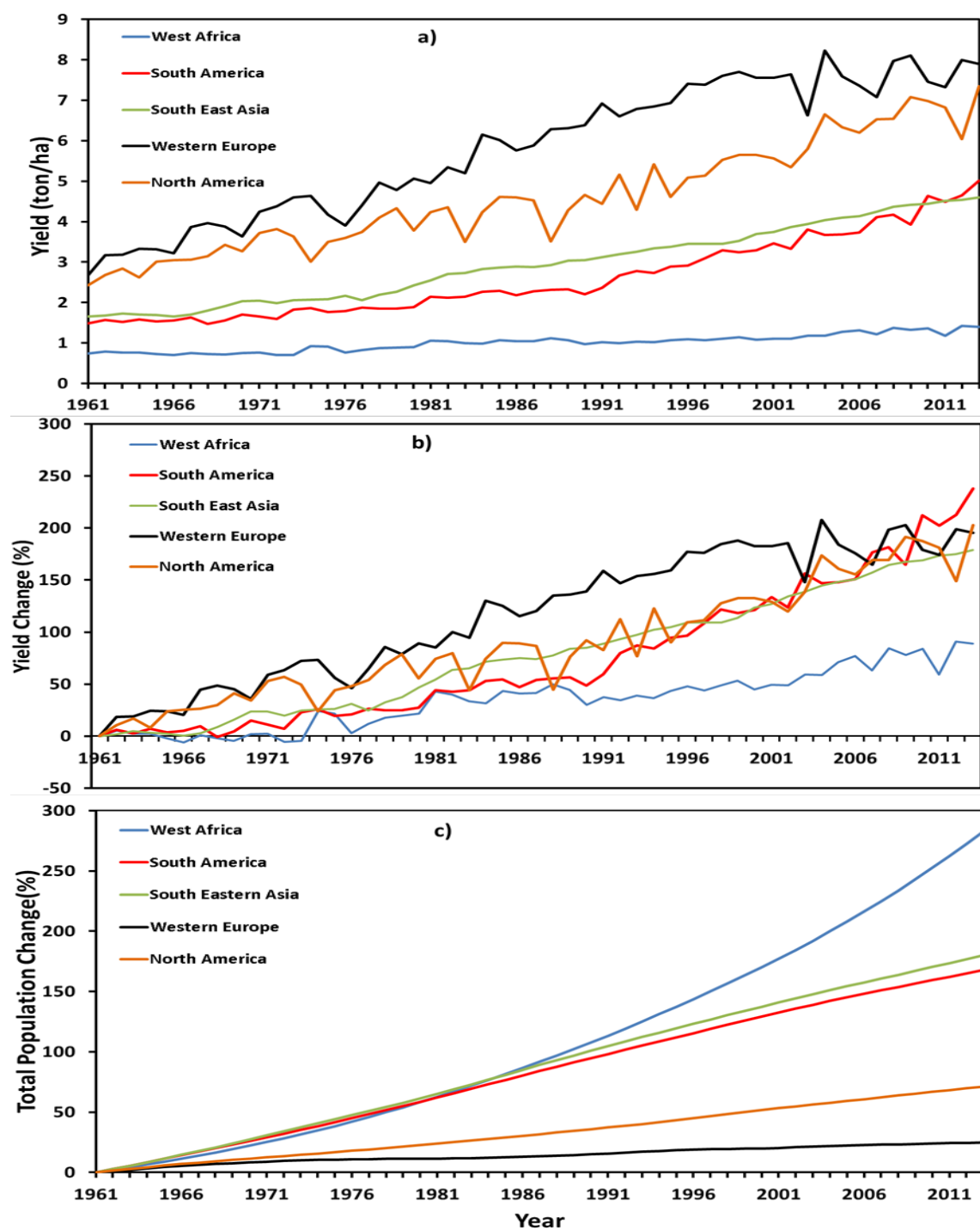


Figure S1: Cereals yield (a, b) and population(c) trend comparison. Source: FAOSTAT, 2017

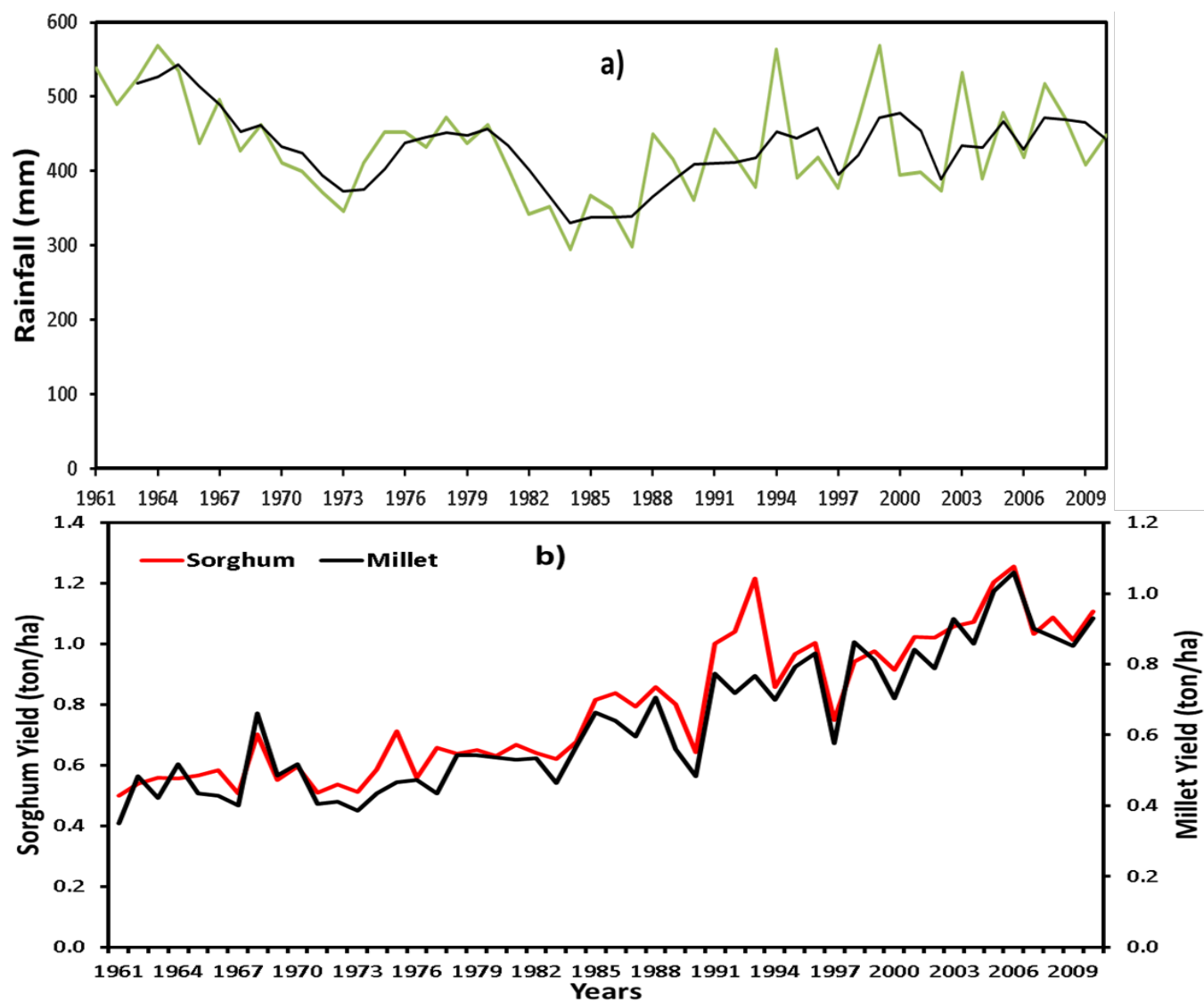


Figure S2. Decreasing rainfall (a) and increasing yields (b) at Dori in Burkina Faso

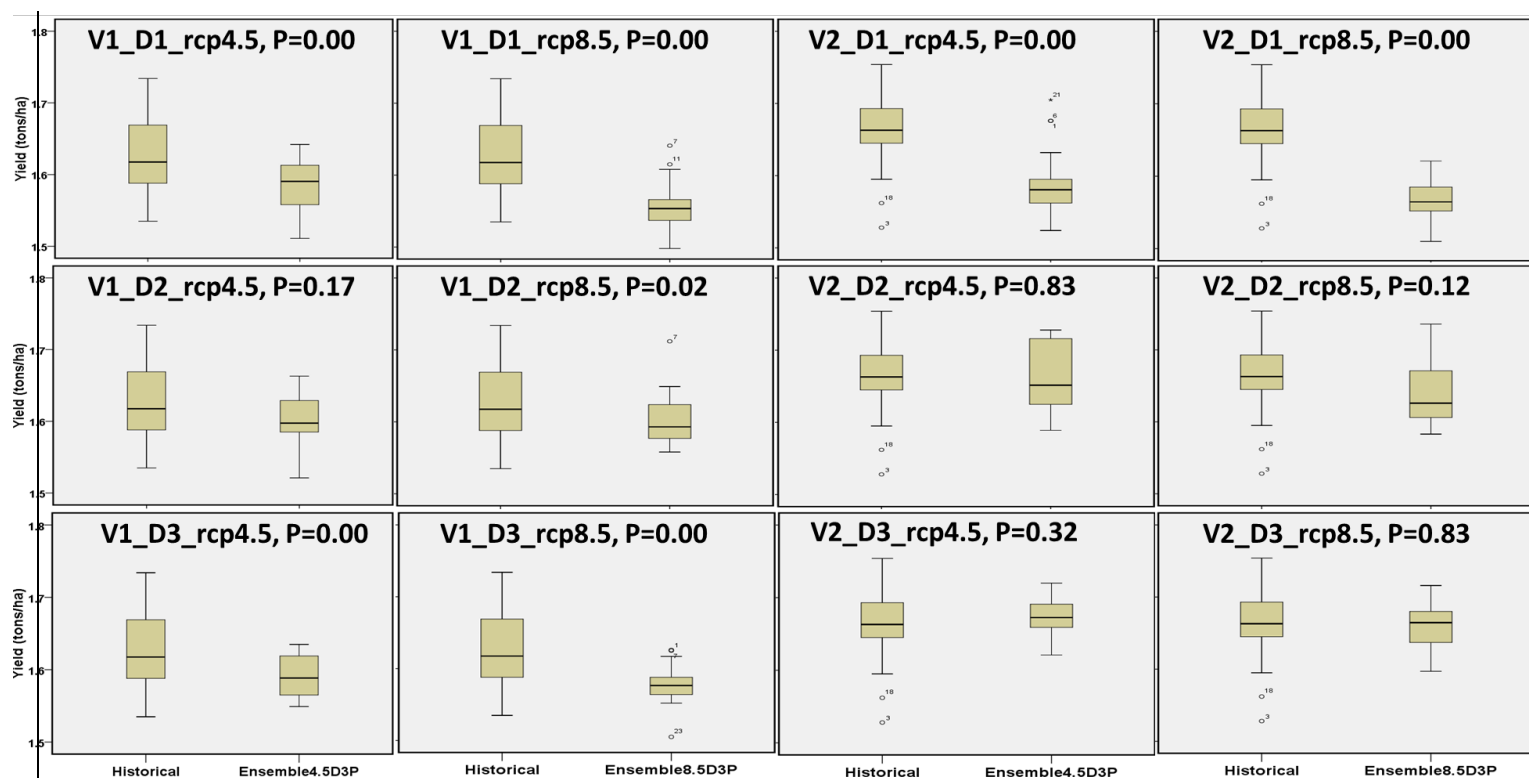


Figure S3: NTarla sorghum yield ensemble (9 GCM models) under the rcp4.5 and 8.5 for future period, 2021/25-2050 relative to the baseline periods, 1981/85-2010.

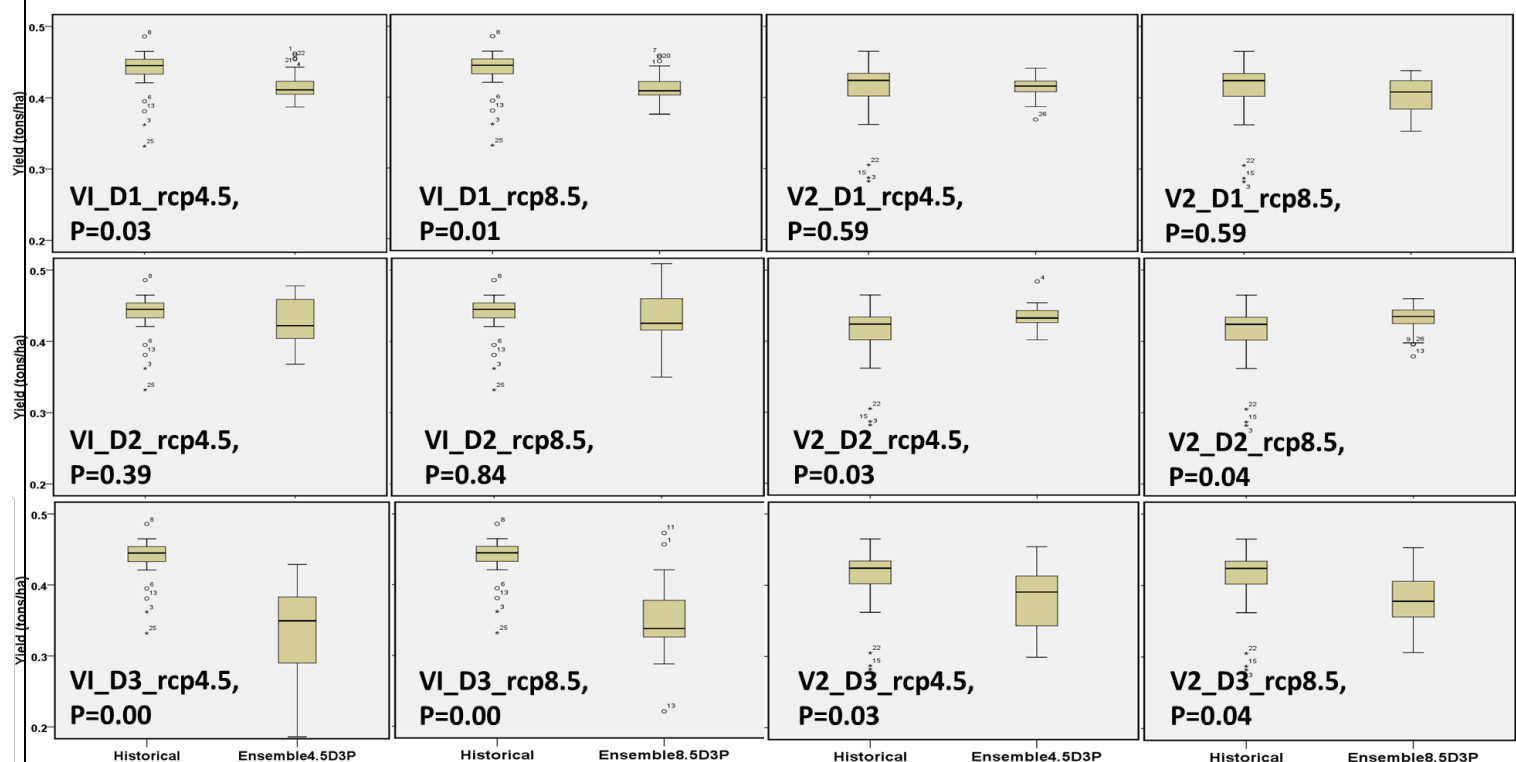


Figure S4: Tillabery millet yield ensemble (9 GCM models) under the rcp4.5 and 8.5 for future period, 2021/25-2050 relative to the baseline period, 1981/85-2010.

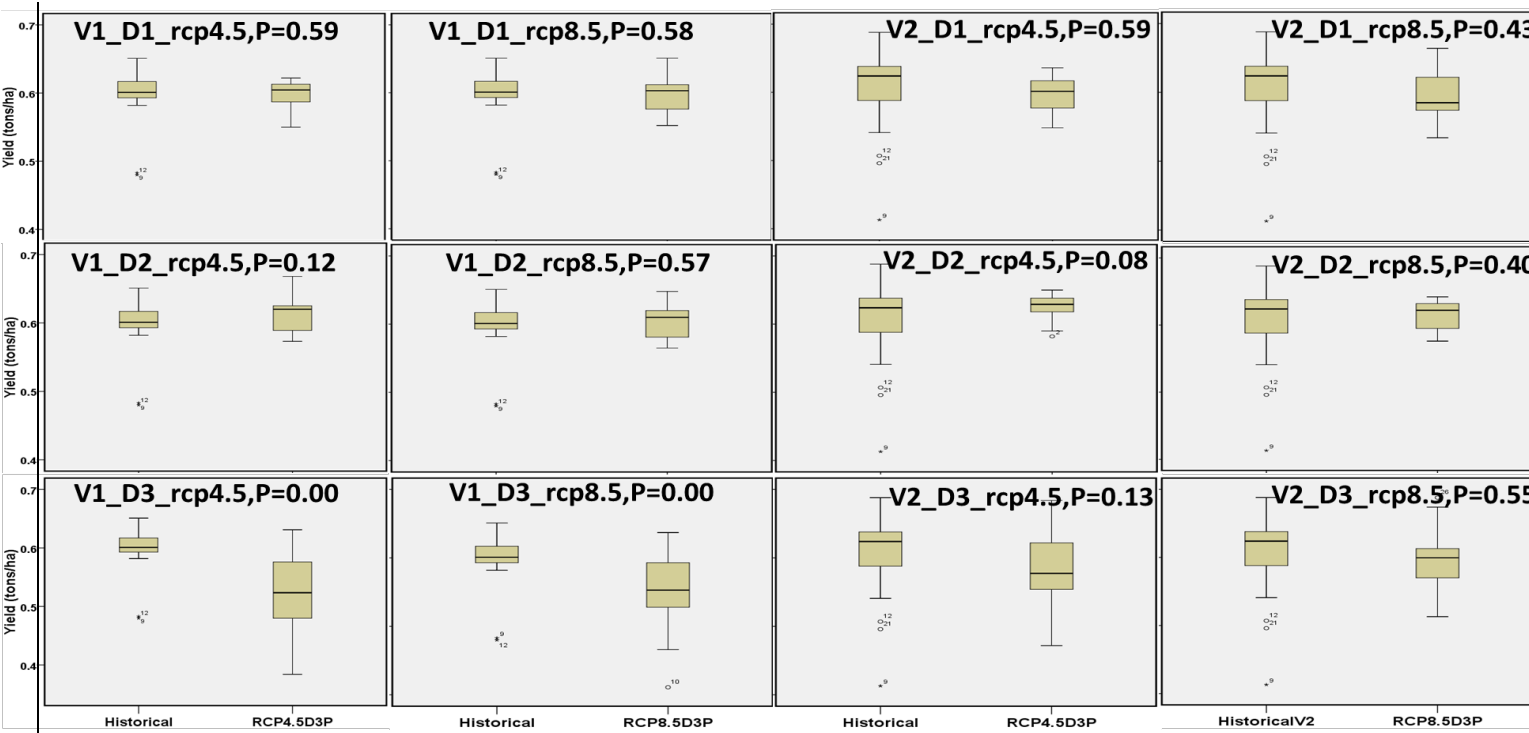


Figure S5: Dori millet yield ensemble (9 GCM models) under the rcp4.5 and 8.5 for future period, 2021/25-2050 relative to the baseline period, 1981/85-2010.

Supplementary Tables.

Table S1: Southern Guinea Savanna (Makurdi) average simulated maize and sorghum grain yield change for the future period (2025-2050) relative to the current period (1985-2010)

Scenarios	Yield(tons/ha)	YΔ(%)	Yield(tons/ha)	YΔ(%)	Yield(tons/ha)	YΔ(%)	Yield(tons/h
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Office for Nuclear Regulation

	<u>MaizeV1</u>		<u>MaizeV2</u>		<u>SorghumV1</u>		<u>SorghumV2</u>
<u>Histv1D1p</u>	<u>1.281</u>	<u>0</u>	<u>1.278</u>	<u>0</u>	<u>1.565</u>	<u>0</u>	<u>1.602</u>
<u>Rcp4.5V1D1P</u>				<u>1.307</u>		<u>2.1</u>	<u>1.329</u>
<u>RCP4.5V1D2P</u>				<u>1.313</u>		<u>2.5</u>	<u>1.340</u>
<u>RCP4.5V1D3P</u>				<u>1.329</u>		<u>3.7</u>	<u>1.344</u>
<u>RCP8.5V1D1P</u>				<u>1.317</u>		<u>3.3</u>	<u>1.335</u>
<u>RCP8.5V1D2P</u>				<u>1.320</u>		<u>3.0</u>	<u>1.347</u>
<u>RCP8.5V1D3P</u>				<u>1.323</u>		<u>3.3</u>	<u>1.337</u>
<u>RCP8.5V1D1M</u>				<u>2.017</u>		<u>57.5</u>	<u>2.033</u>
<u>RCP8.5V1D2M</u>				<u>2.015</u>		<u>57.3</u>	<u>2.050</u>
<u>RCP8.5V1D3M</u>				<u>2.029</u>		<u>58.4</u>	<u>2.049</u>
<u>RCP8.5V1D1NP</u>				<u>2.791</u>		<u>117.9</u>	<u>2.832</u>
<u>RCP8.5V1D2NP</u>				<u>2.796</u>		<u>118.3</u>	<u>2.839</u>
<u>RCP8.5V1D3NP</u>				<u>2.804</u>		<u>118.9</u>	<u>2.834</u>
<u>RCP8.5V1D1OP</u>				<u>3.613</u>		<u>182.1</u>	<u>3.645</u>
<u>RCP8.5V1D2OP</u>				<u>3.590</u>		<u>180.3</u>	<u>3.660</u>
<u>RCP8.5V1D3OP</u>				<u>3.601</u>		<u>181.2</u>	<u>3.615</u>

Note: All are significant at 0.05 except those specified as ^a

Table S2: Northern Guinea Savanna (Samaru illustrated) average simulated maize and sorghum grain yield change for the future period (2021-2050) relative to the current period (1981-2010)

~~ONR site report~~

~~EDF Energy -- Dungeness B~~

~~Report for period: 1 January -- 31 March 2025~~

~~Authored by: Nominated Site Inspector Dungeness B~~

~~Approved by: Joint Head of Regulation -- Operating Reactors~~

~~Issue: 1~~

~~Published: April 2025~~

~~Record reference: 2025/13652~~

Foreword

This report is issued as part of our commitment to make information about inspection and regulatory activities relating to the above site available to the public. Reports are distributed to members for the Dungeness site stakeholder group and are also available on our website: www.onr.org.uk/publications/regulatory-reports/site-specific-reports/llessg-reports.

Our site inspectors usually attend Dungeness site stakeholder group meetings where these reports are presented and will respond to any questions raised there. Any person wishing to inquire about matters covered by this report should contact us via email at contact@onr.gov.uk.

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~~1. Inspections~~

~~1.1. Date(s) of inspection~~

Our site inspector made inspections on the following dates during the report period 01 January—31 March 2025:

~~4—6 February~~

~~14 February~~

~~5—6 March~~

~~2. Routine matters~~

~~2.1. Inspections~~

Inspections are undertaken as part of the process for monitoring compliance with:

- ~~• The conditions attached by ONR to the nuclear site licence granted under the Nuclear Installations Act 1965 (NIA65) (as amended);~~
- ~~• The Energy Act 2013;~~
- ~~• The Health and Safety at Work etc Act 1974 (HSWA74); and~~
- ~~• Regulations made under HSWA74, for example, the Ionising Radiations Regulations 2017 (IRR17) and the Management of Health and Safety at Work Regulations 1999 (MHSWR99).~~

The inspections entail monitoring the licensee's actions on the site in relation to incidents, operations, maintenance, projects, modifications, safety case changes and any other matters that may affect safety. The licensee is required to make and implement adequate arrangements under the conditions attached to the licence in order to ensure legal compliance. Inspections seek to judge both the adequacy of these arrangements and their implementation.

In this period, routine inspections of Dungeness B covered the following:

- ~~• modifications to plant, equipment and safety cases~~
- ~~• radiological protection~~
- ~~• conventional (non-nuclear) health and safety~~
- ~~• security~~

Members of the public who would like further information on our inspection activities during the reporting period can view site inspection records on our website:

www.onr.org.uk/publications/regulatory-reports/site-specific-reports/inspection-records.

Should you have any queries regarding our inspection activities, please email contact@onr.gov.uk.

2.2. Other work

The site inspector met with site safety representatives, to support their function of representing employees and receiving information on matters affecting their health, safety and welfare at work.

The site inspector met regularly with the site-based Independent Nuclear Assurance (INA) team to ensure the internal regulator function remains effective and verifying information provided by the station.

In addition, the site inspector undertook a number of intelligence gathering activities and plant visits including engagement with members of the site's senior leadership team.

3. ~~Non-routine matters~~

~~Licensees are required to have arrangements to respond to non-routine matters and events. Our inspectors judge the adequacy of the licensee's response, including actions taken to implement any necessary improvements.~~

~~There were no such matters or events of significance during the period.~~

4. Regulatory activity

We may issue formal documents to ensure compliance with regulatory requirements. Under nuclear site licence conditions, we issue regulatory documents, which either permit an activity or require some form of action to be taken. These are usually collectively termed licence instruments but can take other forms. In addition, inspectors may take a range of enforcement actions, to include issuing an enforcement notice.

- The following licence instruments, enforcement notices and enforcement letters have been issued during the period:

Table 1: Licence instruments and enforcement notices issued by ONR during this period

ScenariosDate	Yield(tons/ha) MaizeV1Type	YΔ(%)Reference	Yield(tons/ha) MaizeV2Description	YΔ(%)	Yield(tons/ha) SorghumV1	YΔ(%)	Yield(t
Histv1D1p13/03/25	1.628Approval	0LI 569	1.628Approval of the Dungeness B Nuclear Power Station Defueling Emergency Plan	0	1.360	0	1.397
Rcp4.5V1D1P	1.325	-18.6	1.393		-14.4	1.246	
RCP4.5V1D2P	1.341	-17.6	1.348		-17.2	1.278	
RCP4.5V1D3P	1.364	-16.2	1.385		-14.9	1.267	
RCP8.5V1D1P	1.302	-20.0	1.303		-19.9	1.208	
RCP8.5V1D2P	1.319	-19.0	1.321		-18.9	1.238	
RCP8.5V1D3P	1.360	-16.5	1.368		-15.9	1.257	
RCP8.5V1D1M	1.993	22.4	2.014		23.7	1.924	
RCP8.5V1D2M	2.026	24.5	2.032		24.8	1.974	
RCP8.5V1D3M	2.082	27.9	2.103		29.2	1.990	
RCP8.5V1D1NP	2.758	69.4	2.801		72.0	2.724	
RCP8.5V1D2NP	2.790	71.4	2.797		71.8	2.792	
RCP8.5V1D3NP	2.908	78.6	2.921		79.4	2.835	
RCP8.5V1D1OP	3.525	116.5	3.592		120.6	3.687	
RCP8.5V1D2OP	3.595	120.8	3.597		121.0	3.577	
RCP8.5V1D3OP	3.764	131.2	3.801		133.5	3.821	

Note: All are significant at 0.05 level

Tablee S3: Sahelian zone (Dori and Tahoua illustrated) average simulated millet grain yield change for the future period (2021-2050) relative to the current period (1981-2010)

<u>Dori/ Scenarios</u>	<u>Yield(tons/ha) V1</u>	<u>YΔ(%)</u>	<u>Yield(tons/ha) V2</u>	<u>YΔ(%)</u>	<u>Tahoua/ Scenarios</u>	<u>Yield(tons/ha) V1</u>	<u>YΔ(%)</u>	<u>Yield(tons/ha) V2</u>
Histv1D1p	0.601	0	0.604	0	Histv1D1p	0.412	0	0.402
Rcp4.5V1D1P	0.597	-0.7ab	0.598	-1.0ab	Rcp4.5V1D1P	0.411	-0.1ab	0.410
RCP4.5V1D2P	0.614	2.2ab	0.624	3.4	RCP4.5V1D2P	0.417	1.2ab	0.432
RCP4.5V1D3P	0.524	-12.9	0.582	-3.6ab	RCP4.5V1D3P	0.333	-19.2	0.379
RCP8.5V1D1P	0.596	-0.8ab	0.594	-1.6ab	RCP8.5V1D1P	0.400	-2.8ab	0.409
RCP8.5V1D2P	0.606	0.8ab	0.614	1.6ab	RCP8.5V1D2P	0.415	0.7ab	0.420
RCP8.5V1D3P	0.553	-7.9	0.596	-1.3ab	RCP8.5V1D3P	0.353	-14.1	0.395
RCP8.5V1D1M	0.922	53.4	0.871	44.2	RCP8.5V1D1M	0.612	48.7	0.602
RCP8.5V1D2M	0.933	55.3	0.887	46.9	RCP8.5V1D2M	0.633	53.7	0.620
RCP8.5V1D3M	0.687	14.4	0.793	31.3	RCP8.5V1D3M	0.603	46.5	0.480
RCP8.5V1D1NP	1.259	109.5	1.219	101.9	RCP8.5V1D1NP	0.832	102.2	0.831
RCP8.5V1D2NP	1.275	112.1	1.246	106.3	RCP8.5V1D2NP	0.855	107.9	0.846
RCP8.5V1D3NP	0.721	20.0	0.968	60.2	RCP8.5V1D3NP	0.739	79.6	0.495
RCP8.5V1D1OP	1.601	166.4	1.524	152.4	RCP8.5V1D1OP	1.048	154.8	1.017
RCP8.5V1D2OP	1.607	167.3	1.610	166.6	RCP8.5V1D2OP	1.062	158.1	1.069
RCP8.5V1D3OP	0.746	24.1	1.109	83.6	RCP8.5V1D3OP	0.824	100.3	0.495

Note: ab indicate not Significant at 0.05

Reports detailing the above regulatory decisions can be found on our website:
www.onr.org.uk/publications/regulatory-reports/site-specific-reports/project-assessment-reports

ONR issued a routine licence instrument to approve a revision to the Dungeness B emergency plan. This change included moving the map of the detailed emergency planning zone from the emergency plan to the emergency handbook. ONR's assessment confirmed that the revised emergency plan is consistent with the generic emergency plan for a defueling station and that the detailed emergency planning zone contained in the emergency handbook is consistent with that determined by the local authority.

~~5. News from ONR~~

~~For the latest updates and information on our work, please subscribe to our regular email newsletter, ONR News, at www.onr.org.uk/news/newsletter.~~

~~6. Contacts~~

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References

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